

IBM z17 overview

Advanced AI where it matters most

Client presentation

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Mainframe is crucial
in the world today

70%

of all financial transactions
go through an IBM mainframe

\$8.5T

payment volume by
value being processed
on mainframe

Only

315ms

of downtime annually.
This is equivalent
to industry-leading
99.999999% uptime

How well are you leveraging
and building on your
existing **transaction
processing** investment?

There's untapped
potential in ...

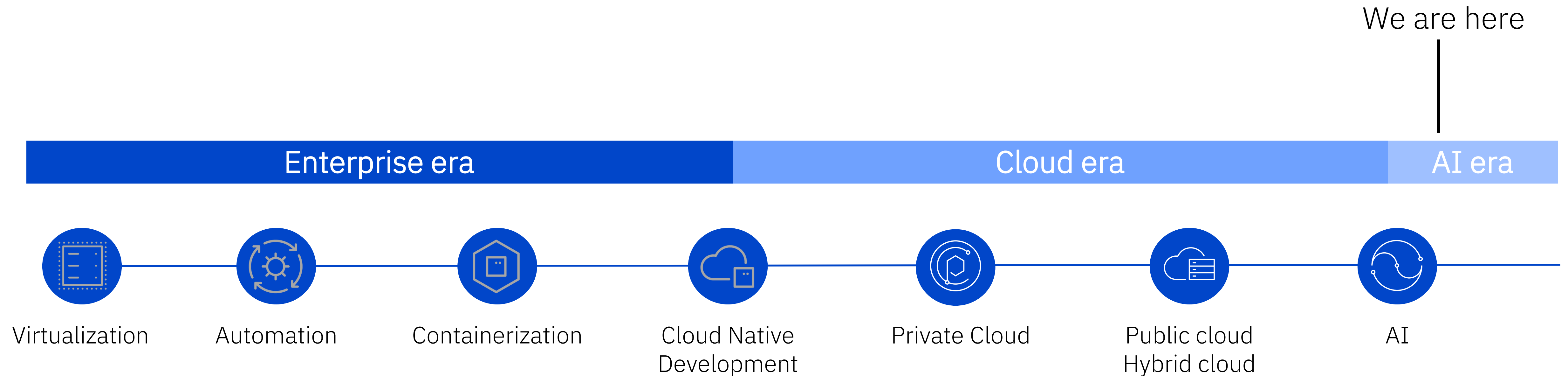
Your **infrastructure**

Your **applications**

Your **data**

The AI revolution is here

AI is the next evolution of what came before



Yet ... AI is placing huge demands on IT

>100x

more model parameters⁴

7x

greater computational throughput⁵

10x

increase in energy consumption⁶

10x

growth in newly-generated AI data⁷

Introducing IBM z17

Enhance your hybrid
cloud with advanced
AI where it matters most



IBM z17 and AI technology will fundamentally
transform [transaction processing and data](#)
to unlock additional productivity and
efficiency for your business

IBM z17

Fully engineered stack for AI where it matters most



Transaction
processing platform

Operating systems
& firmware

IBM Z
infrastructure

Built on a foundation of security,
resiliency, and high availability

450B

Per day with 1ms
response time vs. 300B
on IBM z16

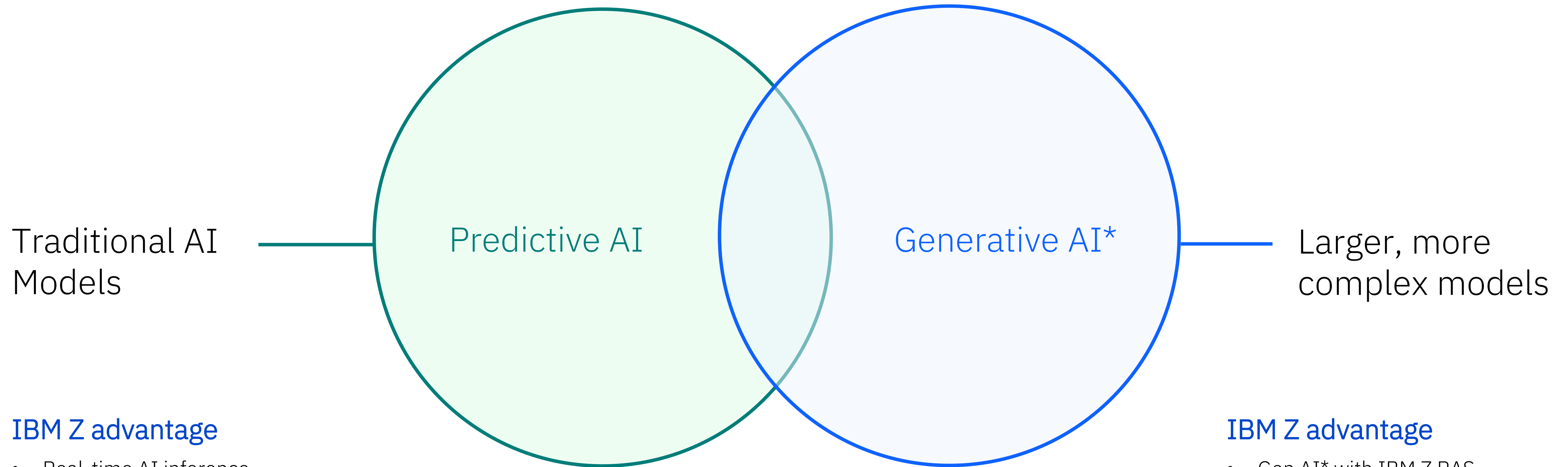
7.5x
AI throughput

Utilizing 8 AI processing
units vs. one on IBM z16

83%
less power
consumption

Moving AI-infused OLTP
workloads from compared
x86 servers

Traditional AI and Gen AI on IBM Z deliver unique capabilities



IBM Z advantage

- Real-time AI inference
- In-transaction AI at scale
- Inference on every transaction¹

IBM Z advantage

- Gen AI* with IBM Z RAS
- Security and data protection
- Energy efficiency

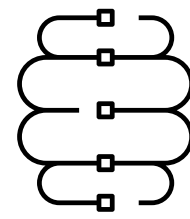
Real-time multiple-model AI only on IBM z17

Generates high value by combining the power of traditional AI models and LLMs in real-time on every transaction which can lead to higher predictive accuracy and expanded insights

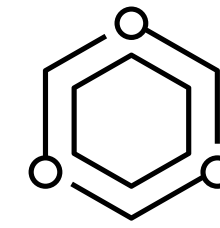
^{*} With IBM Spyre AI Accelerator

Unlock greater AI accuracy, productivity, and agility

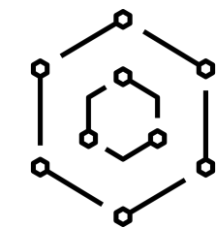
- Apply multiple AI models to transactions for better accuracy with optimal latency
- Increase IT productivity
- Predict and reduce IT incidents for greater resiliency
- On-board new Z skills in half the time



Inferencing



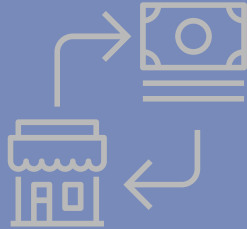
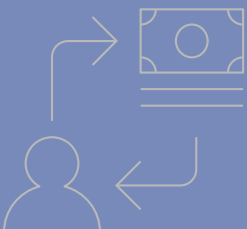
Assistants



Agents

Effective combination of AI inferencing, assistants, and agents* to transform and simplify the mainframe experience*

AI on IBM z17 makes more valuable outcomes possible in every industry

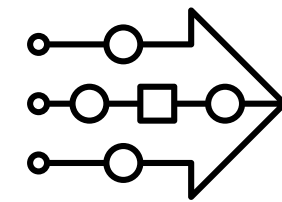
 OVERDRAFT LIMIT DEFAULT	 PAYMENT CONTINGENCY	 FRAUD DETECTION	 CLEARING & SETTLEMENT	 PAYMENTS FRAUD	 PRODUCT SUGGESTION	 ANTI-MONEY LAUNDERING
 LOAN RISK MITIGATION	 GAMING THE SYSTEM	 INSURANCE UNDERWRITING	 CLAIMS FRAUD PREVENTION	 MEDICAL IMAGE RECOGNITION	 GEOSPATIAL ANALYSIS	 DOCUMENT PROCESSING
 RETAIL CRIME PREVENTION	 DATA REDACTION	 CHATBOT SERVICES	 CLIMATE CHANGE IMPACT	 DEV, IT OPERATIONS	 CYBER SECURITY	 SENTIMENT ANALYSIS

IBM z17 makes more possible

Enhance your hybrid cloud with advanced AI where it matters most

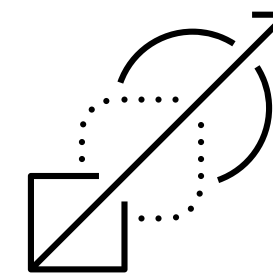


AI for business



Fueling innovation
and growth with AI

Optimizing and Securing your Infrastructure with **AI**



Automating and
transforming
for efficiency



Securing the most
important data

IBM z17

Value of Currency



Price Performance

- **11%** Single thread performance faster response times, improving batch window resiliency
- **12-20%** more capacity within existing footprint to growth within the same frame
- **60%** more system memory capacity to improve online app response time
- Improved software pricing per MIPS
- Next generation of Tailored Fit Pricing for AI and OpenShift® zCX workloads.

Resiliency

- Accelerate your quantum-safe journey with simplified cryptographic inventory and updated quantum-safe APIs based on NIST PQC standards
- Improve productivity and automate tasks with generative AI while keeping your data secure and in place
- Flex Capacity offering, exclusive to IBM z17, for infrastructure testing

Energy Efficiency

- **70%** power reduction on I/O management.
- A typical IBM z17 system can reduce weight by **13%** and is estimated to reduce system power consumption by approximately **17%** compared to a similarly configured IBM z16®.
- A max config IBM z17 system can reduce weight by **26%** and is estimated to reduce system power consumption by approximately **27%** while using **25%** less floor space when compared to a similarly configured IBM z16.

IBM Telum II processor

10 - 36MB L2s

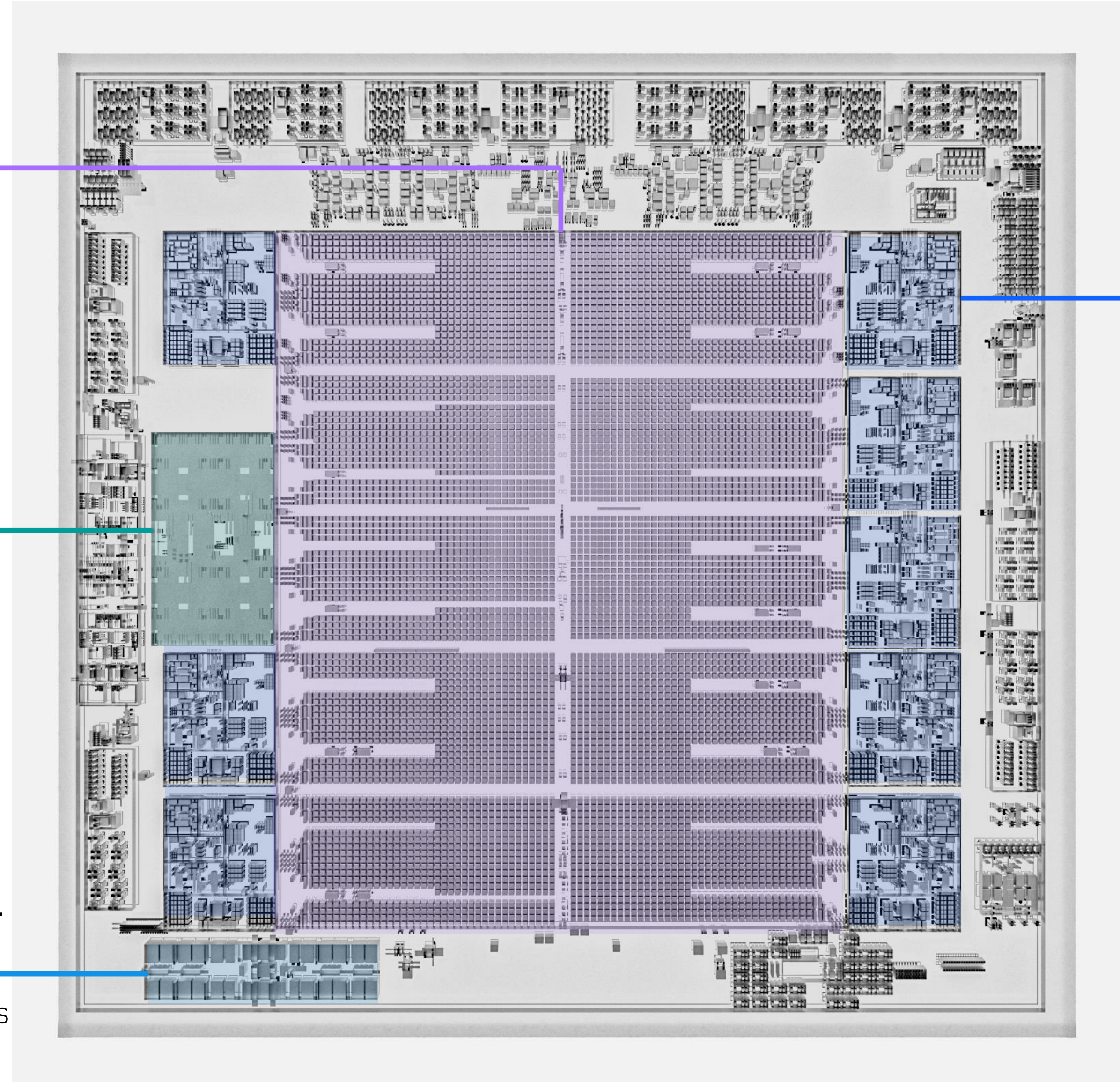
40% more cache per core over z16

I/O DPU

A redesigned I/O subsystem resulting in power and data center footprint reduction

2nd Gen AI accelerator

Improved quantization and matrix operations
8x accelerators available per core



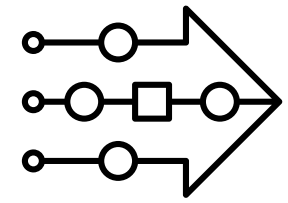
8 - 5.5GHz cores

+11% Single thread performance
20% area reduction and 15%
power reduction

Process up to **450 billion** inference operations per day
with **1 ms** response time, a **50%** increase over z16

IBM z17 makes more possible

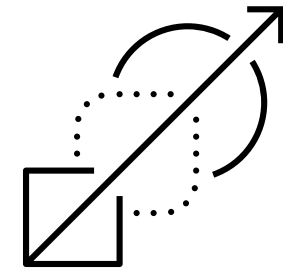
Enhance your hybrid cloud with advanced AI where it matters most



Fueling innovation and growth with AI

- Deliver business growth with deep insights
- Unleash the power of generative AI on data*
- Drive greater impact from your most important data

AI for business



Automating and transforming for efficiency

- Accelerate mainframe application modernization with generative AI*
- Streamline IT Operations with AI insight
- Boost productivity and automate tasks



Securing the most important data

- Proactively identify and mitigate cyber threats
- Quantum-safe protection
- Simplify compliance and improve productivity

Optimizing and Securing your Infrastructure with **AI**

* With IBM Spyre AI Accelerator

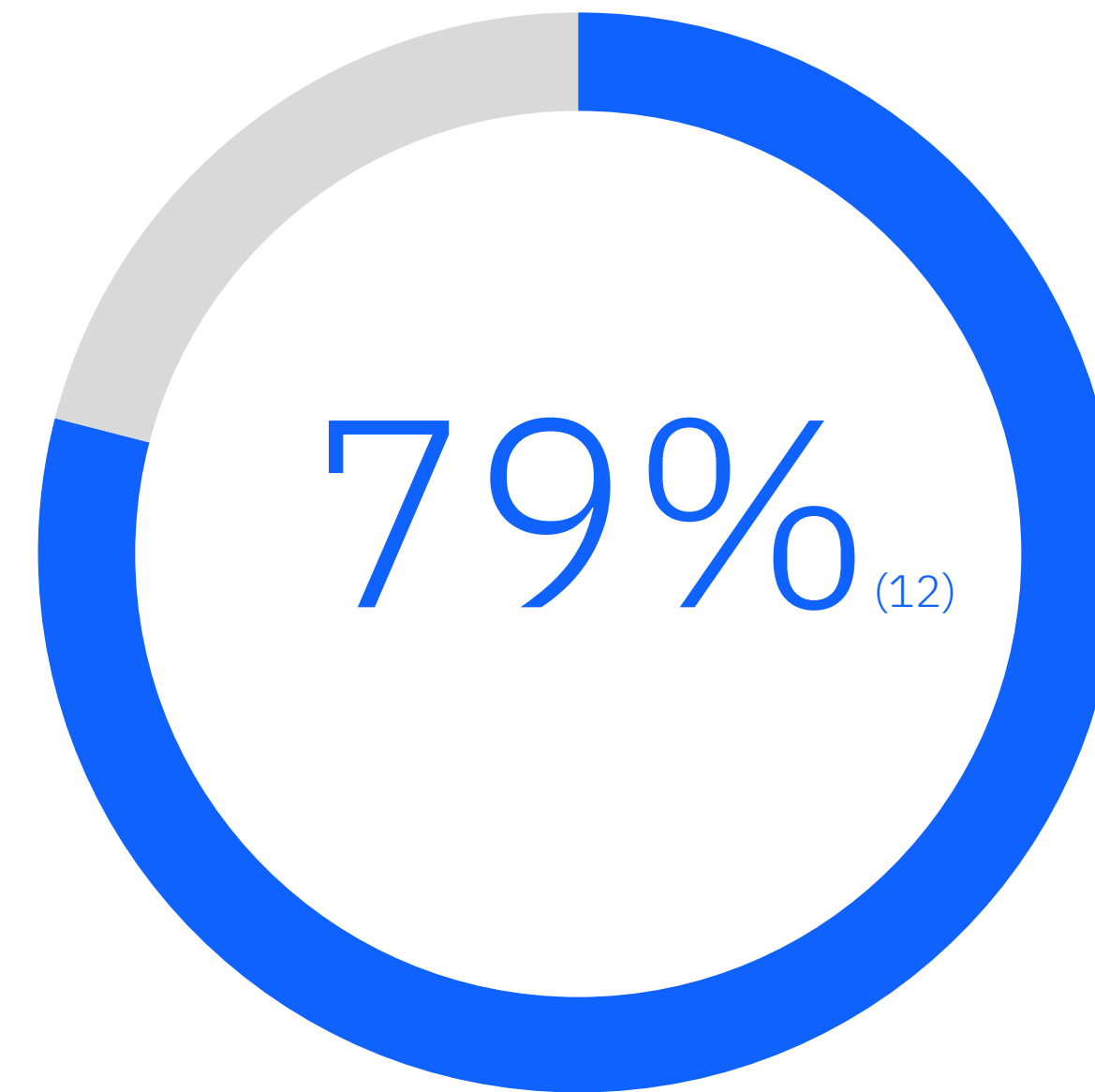
IBM z17 makes more
precision possible:

AI-powered innovation
to fuel business growth

AI adoption is on the rise ... Clients are investing and spending significantly more on AI

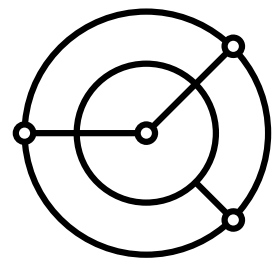
↑ 35%⁽¹¹⁾

Companies are estimated to increase the percentage of their total IT budget spend on AI from 8% to 35% by 2027



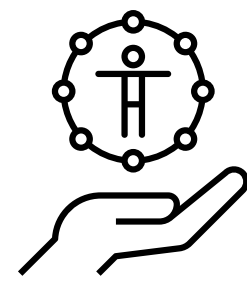
of IT executives agree that mainframes are essential for enabling AI-driven innovation

IBM z17 makes more **precision** possible: AI-powered innovation to fuel business growth



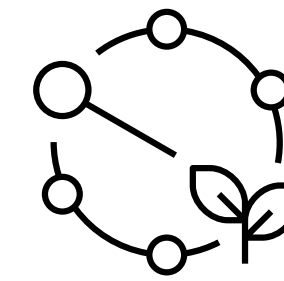
Deliver business growth with deep insights

Improve predictive AI accuracy with advanced inferencing, applying LLMs to mainframe transactions in real-time



Unleash the power of generative AI on your data*

Confidently use your most trusted data with secure generative AI to derive insights by combining transaction context with unstructured data.



Drive greater impact from your most important data

Provide secure, ready access to transaction processing data while maximizing value with analytics and AI

* With IBM Spyre AI Accelerator

IBM z17 makes more
productivity possible:

Automating & transforming
for efficiency

Business efficiency is a top imperative, and mainframe is a key player

78%

of executives agree
that mainframe-based
applications will continue
to have an important role
in digital transformation

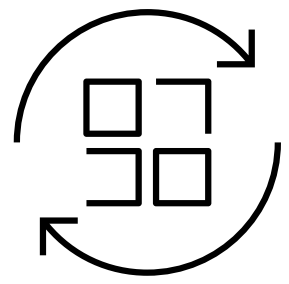
79%

of respondents cited their
top mainframe-related
challenge is acquiring the
right resources and skills
to get work done

61%

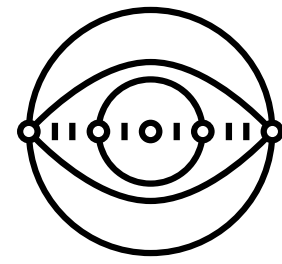
said using gen AI
for app mod efforts
on mainframes is
important to their org

IBM z17 makes more **productivity** possible: Automating and transforming for efficiency



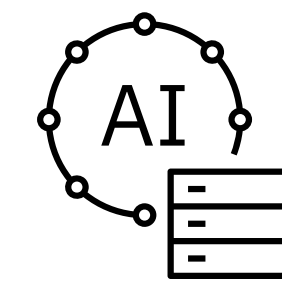
Accelerate application modernization with AI

Increase developer productivity and improve time-to-market with AI infused across the application development lifecycle



Streamline IT operations with AI insight

Leverage AI insight to quickly detect incidents and identify root-cause, minimizing business impact by reducing time to resolution from hours to minutes



Boost productivity and automate tasks

Transform the user experience with AI assistants and agents to guide actions with increased understanding and confidence, reducing learning time for new practitioners

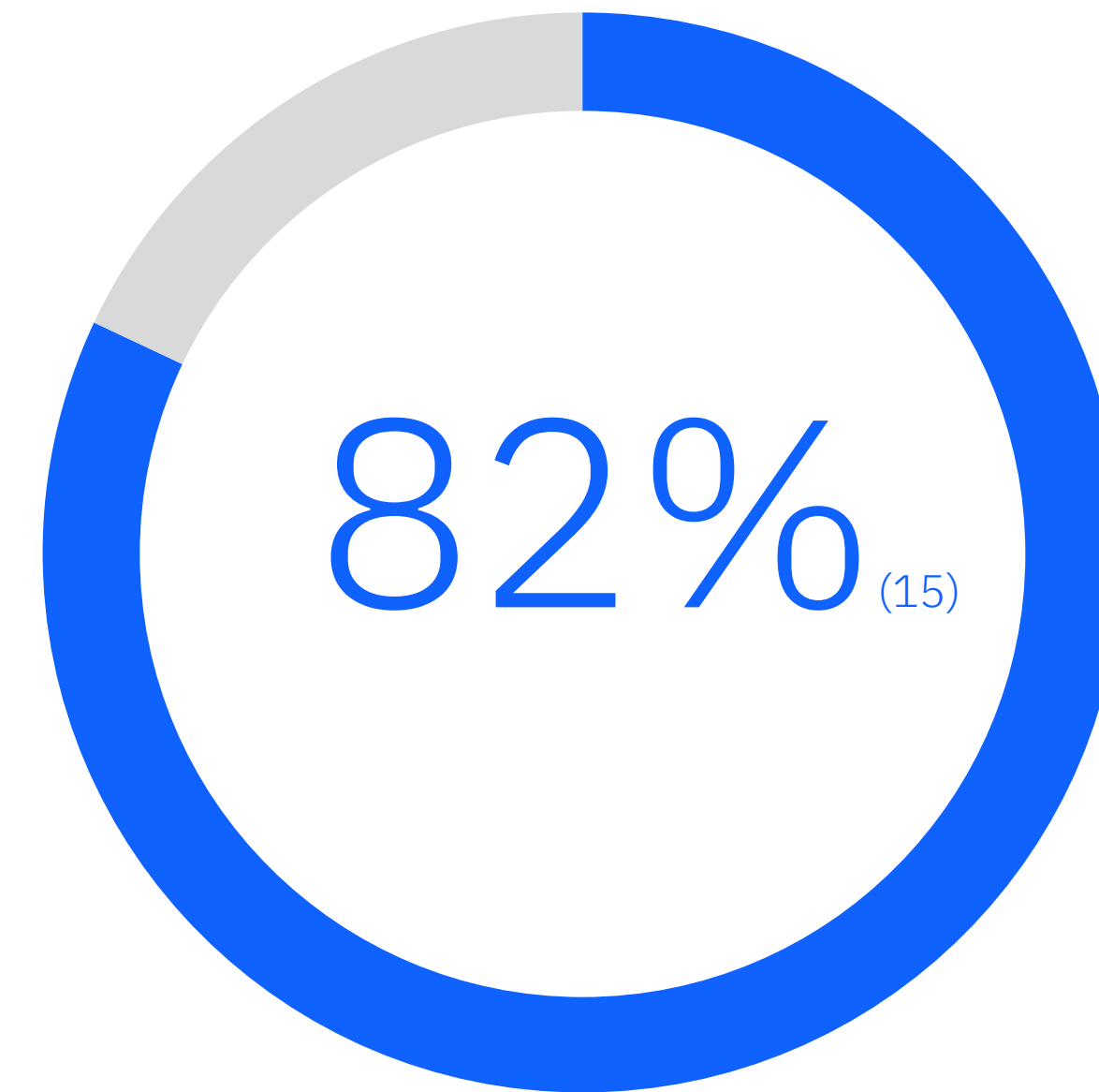
IBM z17 makes
more **security**
possible:

Securing the most
important data

Security threats are everywhere ...
The cost & impact of cyber crime is significant

\$300+M₍₁₄₎

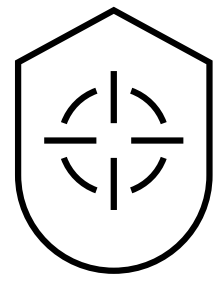
The average cost of a
“mega breach” involving
30-40M records
is over \$300M



of IT executives say that leveraging AI
for monitoring, analyzing, detecting,
and responding to cyber threats is
important to their organizations

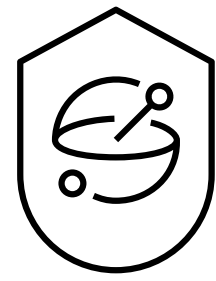
IBM z17 makes more **security** possible:

Securing the most important data



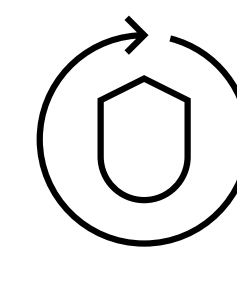
Proactively identify and mitigate cyber threats

Protect and safeguard your reputation and most important data with early threat detection providing the opportunity to mitigate the risk and damage



Quantum-safe protection

Expedite the journey to quantum-safe compliance and protection on the world's first quantum-safe platform



Simplify compliance and improve productivity

Leverage automation to simplify security management tasks and enhance visibility into security vulnerabilities

IBM z17

Client success
and value through AI



IBM CIO office

Leverage mainframe
AI assistant to
transform
operational
experience and
boost productivity
for new users

The IBM CIO Office implementing watsonx Assistant for Z leveraging a ServiceNow and Ansible integration. They prioritized 3 of 35 use cases for initial adoption:

1. Patching Db2 subsystems
2. ServiceNow incident analysis
3. IBM Z Q&A support for operations staff

Went from concept to value in two weeks using on-prem implementation on existing Red Hat OpenShift cluster. Customized authentication and adoption was easy.

Measured operational value for 3 pilot use cases:

Now → Aspirational year-end value

50% → 75%*

reduction of hours to
patch Db2 systems¹

10% → 30%*

reduction of time
to resolution

8% → 30%*

of initial learning time saved,
optimizing usage of SME time

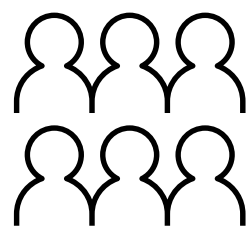
* CIO anticipates that Assistant for Z will accelerate standardization efforts

¹ Assumes a single patch per year with no rollbacks

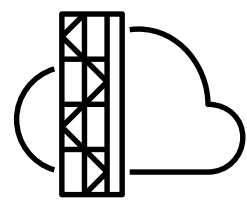
<https://www.ibm.com/case-studies/watsonx-assistant-for-z>

* On-premises on IBM z17
with IBM Spyre AI Accelerator

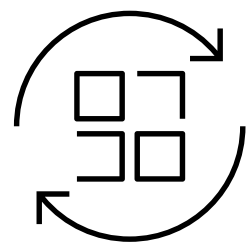
Business value drivers



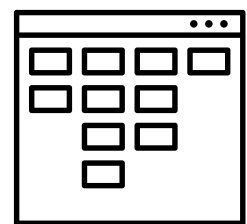
Enhance developer productivity



Increase business agility



Increase operational efficiency



Lower risk

Large European Bank

2-5x
Productivity Gain
in Code Isolation

Financial Company

66%
reduced effort
for understanding
and refactoring

Danske Bank

60% faster
development cycle

Global Logistics Company

14-47% reduced
effort for understanding
& refactoring

58% reduced effort
Code Transformation

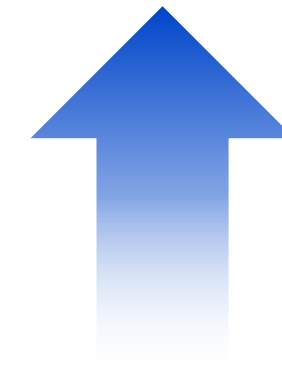
* On-premises on IBM z17
with IBM Spyre AI Accelerator

[Read the Case Study](#)

IBM z17 system highlights

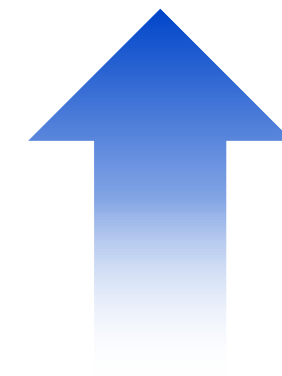


- 1-4 , 19" frames
- 208 c cores
- IBM Telum II processor
- Improved I/O subsystem
- Less weight and less floor space



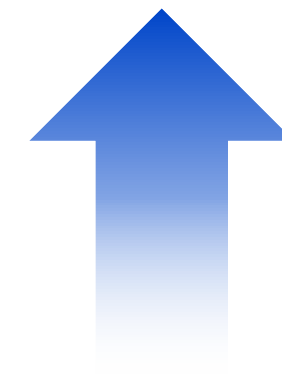
11% Performance

Single thread performance
vs IBM z16



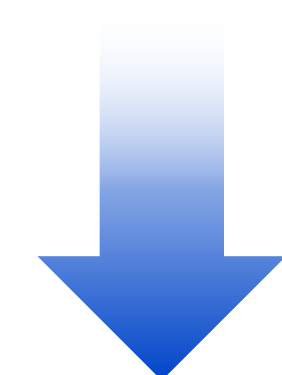
12-20%
Capacity growth

vs IBM z16



60% Memory

Up to 64 TB of memory



17-27% Power

Less power than IBM z16

IBM z17 hardware differentiation

	z16	z17	z17 differentiation
Processor Chip	7nm technology 5.2G Hz 8 cores per chip 18.8 Miles of wire per chip 22.5B transistors	5nm technology 5.5 GHz 8 cores per chip 24.1 Miles of wire per chip 43B transistors	20% core processor area reduction 15% power reduction +11% single thread performance
Capacity	200 cores	208 cores	+12-20% capacity
Cache	32MB L2 256MB virtual L3 2GB virtual L4	36MB L2 360MB Virtual L3 2.8GB virtual L4	+40% cache growth
Total system memory	4U DDR4 DDIMM 40TB max memory	4U DDR5 DDIMM 64TB max memory	+60% system memory growth
On-chip AI acceleration	Telum I	Telum II	More AI processing per chip Up to 8x on-chip AI processors available per core Support for LLM compute primitives Improved quantization: Int8, FP16 datatypes
Off-chip PCIe Accelerator card	N/A	Spyre accelerator 32 Gen AI-ready cores on extended adapters 75W PCIe gen5 x16 adapter Up to 48 adapters per system	Available only on IBM z17
System I/O	Off-chip IO processing	On-processor chip IO DPU	70% Reduced power for IO management Double density with new FICON Express 32G (4-port) Double density with converged network adapter (RoCE and OSA)

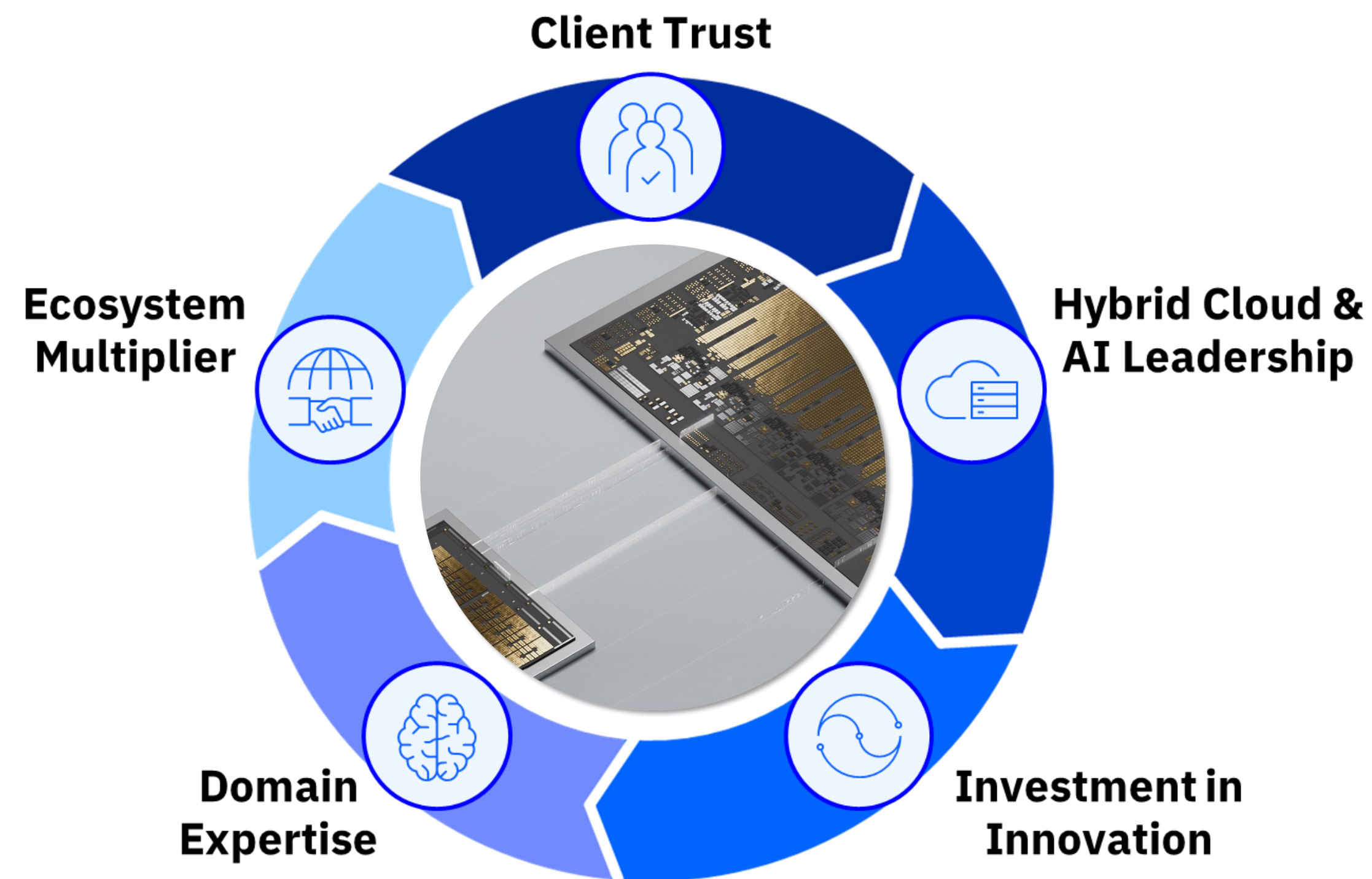
IBM z17

Makes more possible



Possibilities that matter
for your business
for your customers
for the world
making the future better.

Call to action: Do more at the core



Drive IBM Value Creation Framework

Deliver trust and incumbency with resiliency, modernization and automation

Build on IBM Z momentum through innovation to capitalize on mission critical workloads and data gravity

Expand capacity with IBM z17 to address new AI market opportunities

Key takeaways and client next steps

IBM Z is foundational to your customers and the world

IBM z17 with Spyre Accelerator will unlock new paths to value and opportunity

AI is a core strength and differentiator for IBM z17. Adding next-level differentiation to the most secure, available, and resilient transaction processing platform on the planet

Leverage IBM z17 and the watsonx portfolio for Z to address your challenges with skills and complexity. IBM z17 and our transaction processing platform will simplify and streamline operations and development for business efficiency

Set up an AI on IBM Z Discovery workshop with your IBM or business partner sales lead

